DATASCI 406 FA 26, Quiz 2.3

Name	Uniqname	UMID

Each question is worth 1 point. Circle one answer for each.

Question 1 (1 point)

$X \sim$ Poisson(3) and $Y = X \mid X$ is odd. About half of Poisson(3) draws are odd. Which code produces 1000 draws of Y?

1. `x <- rpois(1000, 3); x[x %% 2 == 1]`
2. `x <- rpois(3000, 3); x[x %% 2 == 1][1:1000]`
3. `2 * rpois(1000, 3) + 1`
4. `x <- rpois(1000, 3); x[x %% 2 == 0] <- x[x %% 2 == 0] + 1; x`

Question 2 (1 point)

You are using accept-reject to draw from the half-normal distribution, $f(x) = 2\phi(x)$ for $x \geq 0$ (the target). The candidates come from Exp(1), and $c = \sqrt{2e/\pi}$:

```
f <- function(x) 2 * dnorm(x)
g <- function(x) dexp(x)
c <- sqrt(2 * exp(1) / pi)
```

Which line of code draws 1000 candidates and keeps draws from f?

1. `y <- runif(1000); w <- runif(1000); y[w < f(y) / (c * g(y))]`
2. `y <- rexp(1000); w <- rexp(1000); y[w < f(y) / (c * g(y))]`
3. `y <- rexp(1000); w <- runif(1); y[w < f(y) / (c * g(y))]`
4. `y <- rexp(1000); w <- runif(1000); y[w < f(y) / (c * g(y))]`

Question 3 (1 point)

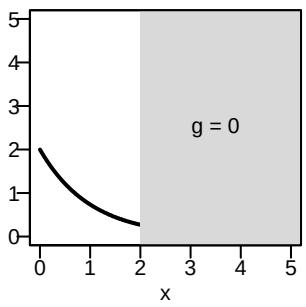
Again using accept-reject, the target is $f(x) = 2x$ on $[0, 1]$ and the candidate is $g(x) = 1$ on $[0, 1]$. What is the smallest valid c, and about what fraction of candidates is accepted?

1. $c = 1$, and every candidate is accepted.
2. $c = 2$, and about half are accepted.
3. $c = 2$, and about two thirds are accepted.
4. $c = 1/2$, and about half are accepted.

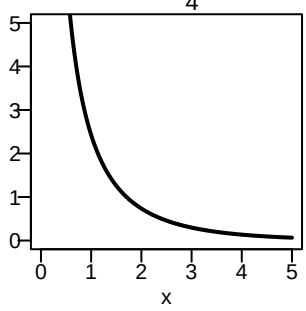
Question 4 (1 point)

Again using accept-reject, the target is Exp(1), $f(x) = e^{-x}$ for $x \geq 0$. Each panel shows the ratio $f(x)/g(x)$ for a candidate density g. Which candidate can be used to draw from f? Circle the panel number.

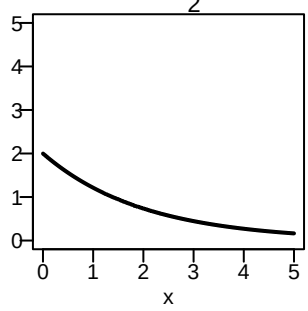
1. $g = U(0, 2)$



2. $g(x) = \frac{x}{4} e^{-x/2}$



3. $g(x) = \frac{1}{2} e^{-x/2}$



4. $g(x) = \frac{4}{\sqrt{2\pi}} e^{-2x^2}$

